WORKED SOLUTION

STEP III 1988, Question 16

Probability and Statistics • Hypergeometric Distribution

STEP III 1988, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s3-q16

The key idea

Drawing without replacement makes the draws dependent, so working through them one at a time is messy. Two ideas get round this. First, count unordered selections to get the distribution of the number of reds. Second, use symmetry: if we imagine the whole urn being emptied in a random order, every ball is equally likely to be in any given position. Writing X as a sum of indicators X_i then gives the mean and variance with no need for the distribution at all. We only need the covariances between pairs.

Solution

The probability of exactly k reds

Think of the balls as distinguishable. The n balls drawn form a set of size n, and every such set is equally likely, so there are $\binom{M}{n}$ equally likely outcomes. A set with exactly k red balls is made by choosing k of the m reds and $n-k$ of the $M-m$ greens. Hence

$$\mathrm{P}(\text{exactly } k \text{ red}) = \frac{\binom{m}{k}\binom{M-m}{n-k}}{\binom{M}{n}},$$

with the usual convention that $\binom{a}{b} = 0$ if $b < 0$ or $b > a$. (This covers the impossible cases, such as $n-k > M-m$.) This is the *hypergeometric* distribution.

Part (i)

Imagine continuing to draw until the urn is empty. The order in which the M (distinguishable) balls come out is a uniformly random arrangement: all $M!$ orders are equally likely. Stopping after n draws doesn't change the first n positions, so we may work with the full arrangement.

Fix a position i. The number of arrangements with a particular ball in position i is

$(M-1)!$, the same for every ball. So the ball in position i is equally likely to be any of the M balls, and

$$\mathrm{P}(X_i = 1) = \frac{m\,(M-1)!}{M!} = \frac{m}{M},$$

as required.

> **Ask yourself** Surely the second ball's colour depends on what the first ball was?

It does, given the first ball. But we want the *unconditional* probability. For $i = 2$ the law of total probability gives

$$\mathrm{P}(X_2 = 1) = \frac{m}{M}\cdot\frac{m-1}{M-1} + \frac{M-m}{M}\cdot\frac{m}{M-1} = \frac{m(M-1)}{M(M-1)} = \frac{m}{M}.$$

The symmetry argument does the same calculation for every i at once.

Part (ii)

Fix positions $i \neq j$ (so $M \geq 2$). Count the arrangements with red balls in both positions. There are m choices for the ball in position i, then $m-1$ for position j, and $(M-2)!$ ways to arrange the rest. Hence

$$\mathrm{P}(X_i = 1 \text{ and } X_j = 1) = \frac{m(m-1)(M-2)!}{M!} = \frac{m(m-1)}{M(M-1)},$$

as required. In other words, the ordered pair of balls in positions (i, j) is equally likely to be any of the $M(M-1)$ ordered pairs of distinct balls.

Mean and variance of X

X counts the red balls among the first n drawn. By linearity of expectation, which needs no independence, and since $\mathrm{E}(X_i) = \mathrm{P}(X_i = 1)$,

$$\mathrm{E}(X) = \sum_{i=1}^{n} \mathrm{E}(X_i) = \frac{nm}{M}.$$

For the variance, use $\mathrm{Var}(X) = \mathrm{E}(X^2) - [\mathrm{E}(X)]^2$. Expanding the square of the sum, and using $X_i^2 = X_i$ because X_i is 0 or 1,

$$X^2 = \sum_{i=1}^{n} X_i^2 + \sum_{i\neq j} X_i X_j = X + \sum_{i\neq j} X_i X_j,$$

where the second sum runs over the $n(n-1)$ ordered pairs (i, j) with $i \neq j$. Now $X_i X_j = 1$ exactly when both balls are red, so by part (ii) $\mathrm{E}(X_i X_j) = \dfrac{m(m-1)}{M(M-1)}$.

Therefore, for $M \geq 2$,

$$\begin{aligned}\mathrm{Var}(X) &= \frac{nm}{M} + \frac{n(n-1)m(m-1)}{M(M-1)} - \frac{n^2m^2}{M^2} \\ &= \frac{nm}{M^2(M-1)}\Big[M(M-1) + (n-1)(m-1)M - nm(M-1)\Big].\end{aligned}$$

The bracket expands to

$$\begin{aligned}&M^2 - M + nmM - nM - mM + M - nmM + nm \\ &\quad = M^2 - nM - mM + nm = (M-m)(M-n).\end{aligned}$$

Hence

$$\mathrm{Var}(X) = \frac{nm(M-m)(M-n)}{M^2(M-1)}.$$

(If $M = 1$ then $n \leq 1$, there are no pairs, and X is constant, so $\mathrm{Var}(X) = 0$.)

It is illuminating to write $p = m/M$. Then

$$\mathrm{E}(X) = np, \qquad \mathrm{Var}(X) = np(1-p) \cdot \frac{M-n}{M-1}.$$

This is the binomial variance multiplied by a correction factor, which is less than 1 when $n \geq 2$.

Checks

- $n = M$: we take every ball, so $X = m$ for certain. The formula gives mean m and variance 0, because of the factor $M - n$.
- $n = 1$: X is a single Bernoulli variable with $p = m/M$, and the variance reduces to $p(1-p)$.
- $M = 6$, $m = 2$, $n = 3$: the formula gives $\mathrm{Var}(X) = \frac{3 \cdot 2 \cdot 4 \cdot 3}{36 \cdot 5} = \frac{2}{5}$. Enumerating all 6! orders exactly agrees, as does every case with $M \leq 6$.

Common slips

- Treating the X_i as independent and getting the binomial variance $np(1-p)$. They are negatively correlated: a red ball drawn early makes later reds less likely.
- Forgetting that the double sum $\sum_{i \neq j}$ has $n(n-1)$ terms, not $\binom{n}{2}$, when the pairs are ordered.

Beyond STEP

Writing a count as a sum of indicator variables is one of the most powerful tricks in probability. It gives means and variances even when the distribution itself is hopeless to

write down. The factor $\frac{M-n}{M-1}$ is the *finite population correction* used in survey sampling. It shows why a poll of 1000 people is almost as informative about a country as it would be about an infinite population. Blitzstein and Hwang, *Introduction to Probability*, uses indicators throughout and treats the hypergeometric distribution in detail.

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